

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Medium

Question

An insect moves at a speed of $\frac{3}{20}$ feet per second. What is this speed, in yards per second? (**3 feet = 1 yard**)

Answer

- A. $\frac{1}{20}$
- B. $\frac{9}{20}$
- C. 6
- D. 20

Correct Answer: A

Rationale

Choice A is correct. It’s given that **3 feet = 1 yard**. It follows that a speed of $\frac{3}{20}$ feet per second is equivalent to $\left(\frac{\frac{3}{20} \text{ feet}}{1 \text{ second}}\right)\left(\frac{1 \text{ yard}}{3 \text{ feet}}\right)$, which is equivalent to $\left(\frac{3}{20}\right)\left(\frac{1}{3}\right)$, or $\frac{1}{20}$, yards per second.

Choice B is incorrect. This is the speed, in feet per second, that’s equivalent to $\frac{3}{20}$ yards per second.

Choice C is incorrect. This is the speed, in yards per second, that’s equivalent to **18**, not $\frac{3}{20}$, feet per second.

Choice D is incorrect. This is the speed, in yards per second, that’s equivalent to **60**, not $\frac{3}{20}$, feet per second.